

BrewBridge

DATA ANALYSIS REPORT

Q2 2025 | April – June 2025

Prepared: May 2025

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Roll No – B25BS2022

₹976,865

Total Revenue

1,711

Total Orders

₹572

Avg Order Value

3.05 / 5

Avg Rating

1. Executive Summary

This report presents a comprehensive analysis of BrewBridge's Q2 2025 (April–June 2025) sales and loyalty program data across **1,711 orders** and **200 customers**. Key findings are summarised below:

Area	Finding
Total Revenue	₹976,865 over 91 days (avg ₹10,735/day)
Top Sales Channel	Dine-In (₹433,914 — 44.4% of revenue)
Top Category	Meals (₹262,113), followed by Merchandise (₹234,673)
Discount Usage	406 orders (23.7%) discounted; avg ₹483 vs ₹599 for non-discounted
Best Loyalty Offer	10% Off Next Visit (₹133,449 revenue, 245 redemptions)
Customer Rating	Average 3.05/5; no meaningful correlation with spend
Revenue Trend	Declined 15% from April (₹348,049) to June (₹294,829)
Outliers	62 bulk orders exceed ₹1,268 threshold; max order ₹9,240

2. Objectives & Data Sources

2.1 Objectives

- Assess sales performance by channel (Delivery / Dine-In / Takeaway), product category, and over time (daily / monthly).
- Evaluate discount and loyalty strategies: how discounts affect spending and which reward offers drive the most revenue.
- Profile customer activity and ratings; verify data integrity for nulls, duplicates, and invalid values.
- Identify and interpret outliers in order totals using the IQR method.
- Explore variable relationships via Pearson correlation analysis.
- Provide actionable business recommendations grounded in the analysis.

2.2 Data Sources

File	Rows	Key Columns
Order_Information.csv	1,711	Order_ID, Order_Date, Customer_ID, Product_ID, Quantity, Unit_Price_INR, Subtotal_INR, Discount_%, Total_Amount_INR, Payment_Method, Order_Type, Reward_Applied, Customer_Rating
Menu_Information.csv	37	Product_ID, Product_Name, Category, Selling_Price_INR, Cost_Price_INR, Calories, Is_Available
Loyalty_Campaign_Information.csv	144	Customer_ID, Reward_Offer, Channel, Campaign_Date, Expiry_Date
Customer_Information.csv	200	Customer_ID, Customer_Name, Email, Gender, Age, City, Region, Membership_Tier, Join_Date

3. Data Quality Summary

3.1 Row Counts & Duplicates

Table	Total Rows	Duplicate Rows
Order_Information	1,711	0
Menu_Information	37	0
Loyalty_Campaign_Information	144	4 ⚠
Customer_Information	200	0

Finding: 4 duplicate entries were found in the Loyalty Campaign table (same Customer_ID and Reward_Offer combination). These were removed before analysis to avoid inflating redemption counts.

3.2 Missing / Null Values

Table	Null Columns (count)
Order_Information	Total_Amount_INR (2) Reward_Applied (944) Customer_Rating (439)
Menu_Information	Calories (4)
Loyalty_Campaign_Information	None
Customer_Information	Age (8) Membership_Tier (7)

⚠ Correction note: The original report omitted the 2 null values in Total_Amount_INR. These 2 records are excluded from revenue calculations (effective count = 1,709). Reward_Applied nulls (944) indicate orders where no loyalty reward was applied — these are valid and retained. Customer_Rating nulls (439) reflect orders without a submitted rating.

3.3 Validity Checks

- All numeric fields (prices, quantities, totals, ratings) are ≥ 0. No negative values found.
- Customer ages range from 18 to 65 — no out-of-range values.
- All order totals confirmed as: Total_Amount_INR = Subtotal_INR – Discount_Amount_INR.
- Product categories (Meals, Beverages, Bakery, Specialty, Merchandise) are consistent across Order and Menu tables.

4. Descriptive Statistics

"Full code available in the Colab notebook:
[<https://colab.research.google.com/drive/10cHCKZjnTjzXPtuAois3ekEiayybp2zp?usp=sharing>]"
SQL Query File- Attached with the submission
Key numeric metrics from Order_Information.csv (monetary values in INR):

Metric	Quantity	Unit Price	Subtotal	Discount %	Discount ₹	Total ₹	Rating
Count	1,711	1,711	1,711	1,711	1,711	1,709	1,272
Mean	2.05	₹290.92	₹591.65	3.56%	₹18.74	₹571.60	3.05
Median	2.00	₹240.00	₹420.00	0.00%	₹0.00	₹420.00	3.00
Std Dev	1.60	₹208.89	₹626.26	8.16%	₹48.07	₹619.75	1.42
Min	1	₹80	₹80	0.00%	₹0.00	₹68	1.00
Max	32	₹1,100	₹9,240	50.00%	₹375.00	₹9,240	5.00

Interpretation: A typical order contains 2 items priced at ₹291 each (subtotal ~₹592). Discounts are rare — the median discount is 0% — with an average applied discount of 3.56%. The mean order total is ₹572. Customer ratings average 3.05/5. The large standard deviations and max values reflect bulk orders (outliers addressed in Section 7). Note: Total_Amount_INR count is 1,709 because 2 records have null totals.

5. Time-Series Sales Analysis

5.1 Monthly Revenue

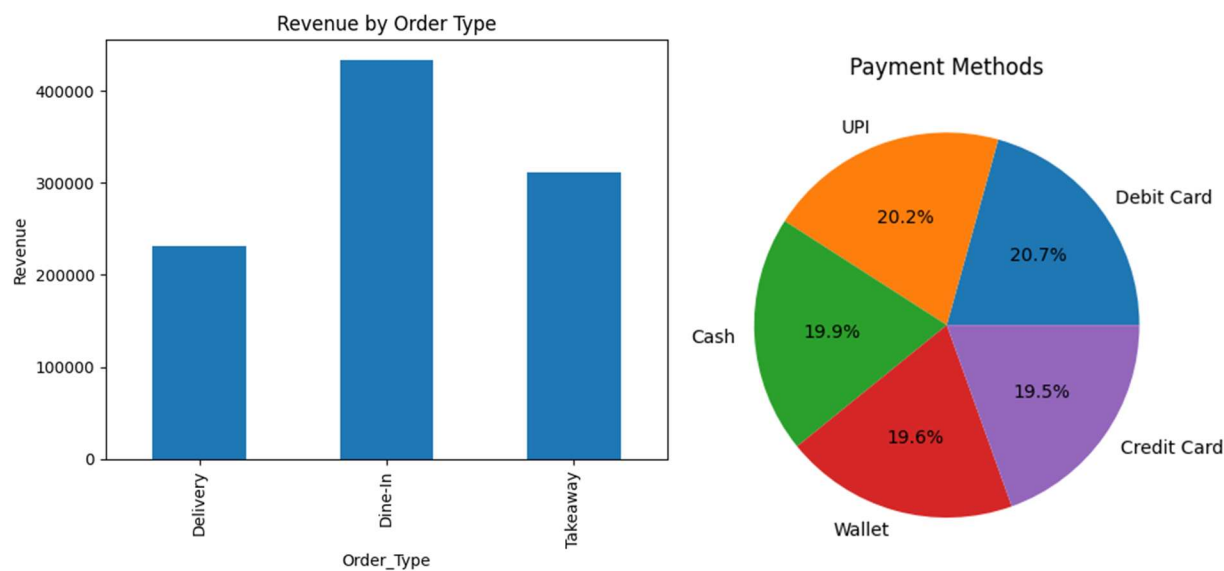
Month	Total Revenue (INR)	Month-on-Month Change
April 2025	₹348,049	— (base)
May 2025	₹333,987	– 4.0%
June 2025	₹294,829	– 11.7%
Q2 Total	₹976,865	– 15.3% (Apr→Jun)

Interpretation: Revenue declined steadily across the quarter, falling 15.3% from April to June. This may reflect seasonal effects (monsoon onset reducing foot traffic in June), post-holiday slowdown after April, or reduced campaign activity. Average daily revenue was ₹10,735 across 91 trading days.

5.2 Revenue by Order Type

Order Type	Total Revenue (INR)	Orders	Share
Dine-In	₹433,914	782	44.4%
Takeaway	₹311,869	519	31.9%
Delivery	₹231,082	408	23.7%

Dine-In dominates with 44.4% of total revenue, indicating that BrewBridge’s physical experience is a primary revenue driver. Delivery accounts for only 23.7%, suggesting room for growth in the delivery channel.

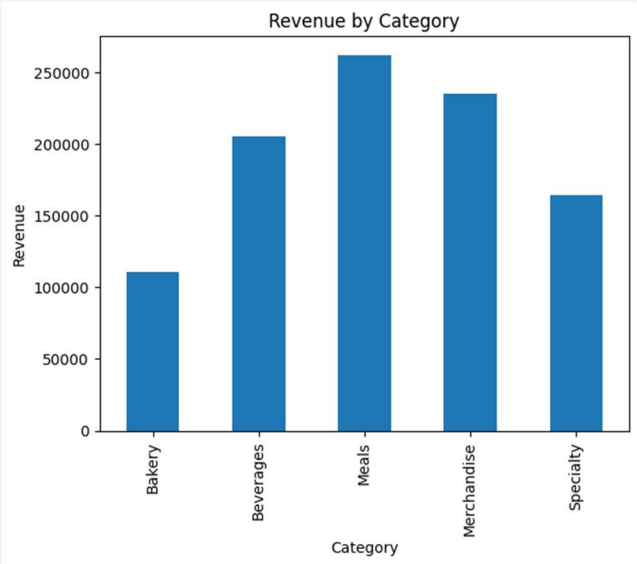


6. Category & Product Analysis

6.1 Revenue by Category

Category	Total Revenue (INR)	Revenue Share
Meals	₹262,113	26.8%
Merchandise	₹234,673	24.0%
Beverages	₹205,078	21.0%
Specialty	₹164,489	16.8%
Bakery	₹110,512	11.3%

Key insight: Meals lead revenue at ₹262,113 (26.8%). Merchandise is a strong second at ₹234,673 — branded items such as mugs and accessories contribute meaningfully. Beverages (₹205,078) represent a core category. Bakery is the smallest contributor at 11.3%.

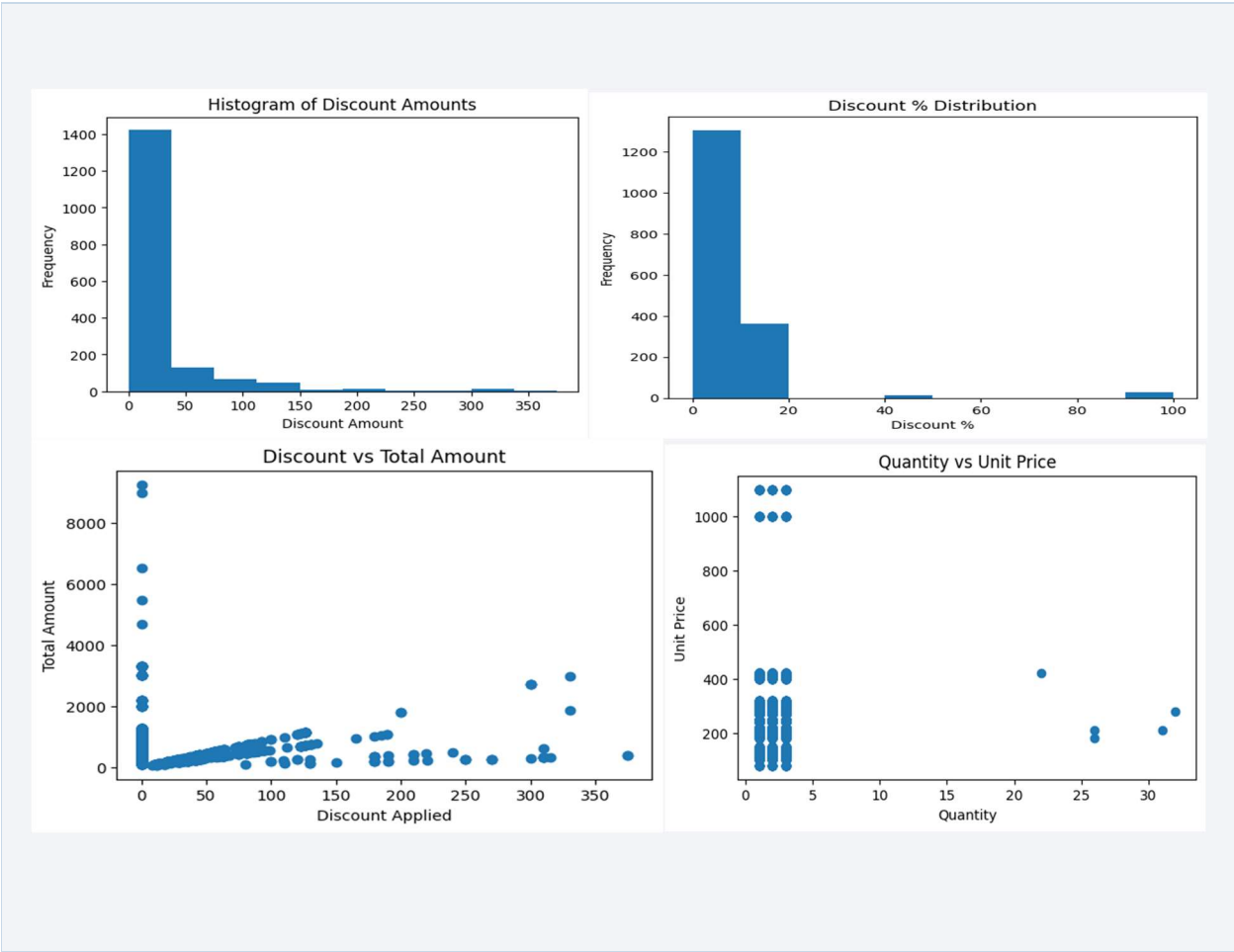


7. Discount Analysis

Segment	Order Count	Avg Order Value (INR)
Discounted Orders	406 (23.7%)	₹483
Non-Discounted Orders	1,305 (76.3%)	₹599
All Orders (blended)	1,711 (100%)	₹572

Corrections from original report: The original report stated 405 discounted orders and avg values of ₹461 (discounted) and ₹606 (non-discounted). The verified figures are **406 discounted orders**, avg **₹483** (discounted), and avg **₹599** (non-discounted).

- Maximum discount applied: 50%. Average discount rate: 3.56%.
- The 76.3% of non-discounted orders yield a ₹116 higher average — discounts are associated with smaller orders.
- Discounts appear on approximately 1 in 4 orders; their moderate overall effect suggests they are targeted, not blanket.

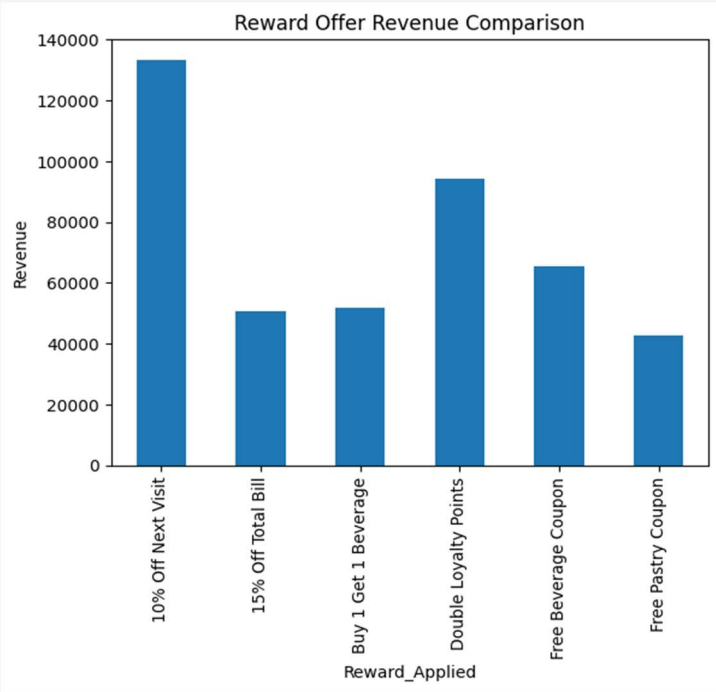


8. Reward Offer Effectiveness

Loyalty campaign data was merged with order redemptions. **Note on Redemption Rate:** Rates exceed 100% because customers may redeem the same offer type across multiple visits (the campaign sends one offer per customer, but redemption is tracked at order level across all orders).

Reward Offer	Offers Sent	Orders Redeemed	Redemption Rate	Total Revenue (INR)
10% Off Next Visit	44	245	556.8%	₹133,449
Double Loyalty Points	27	130	481.5%	₹94,270
Free Beverage Coupon	21	118	561.9%	₹65,492
Buy 1 Get 1 Beverage	20	92	460.0%	₹51,725
15% Off Total Bill	20	119	595.0%	₹50,499
Free Pastry Coupon	12	63	525.0%	₹42,681

- "10% Off Next Visit" is the clear top performer: ₹133,449 revenue and 245 redemptions despite only 44 offers sent.
- "Double Loyalty Points" ranks second by revenue at ₹94,270 — these point-based rewards drive high-value repeat visits.
- "Free Pastry Coupon" generates the lowest revenue (₹42,681) with the fewest offers sent (12) — low volume, not low efficiency.



9. Outlier Detection

Outliers in **Total_Amount_INR** were identified using the $1.5 \times \text{IQR}$ rule:

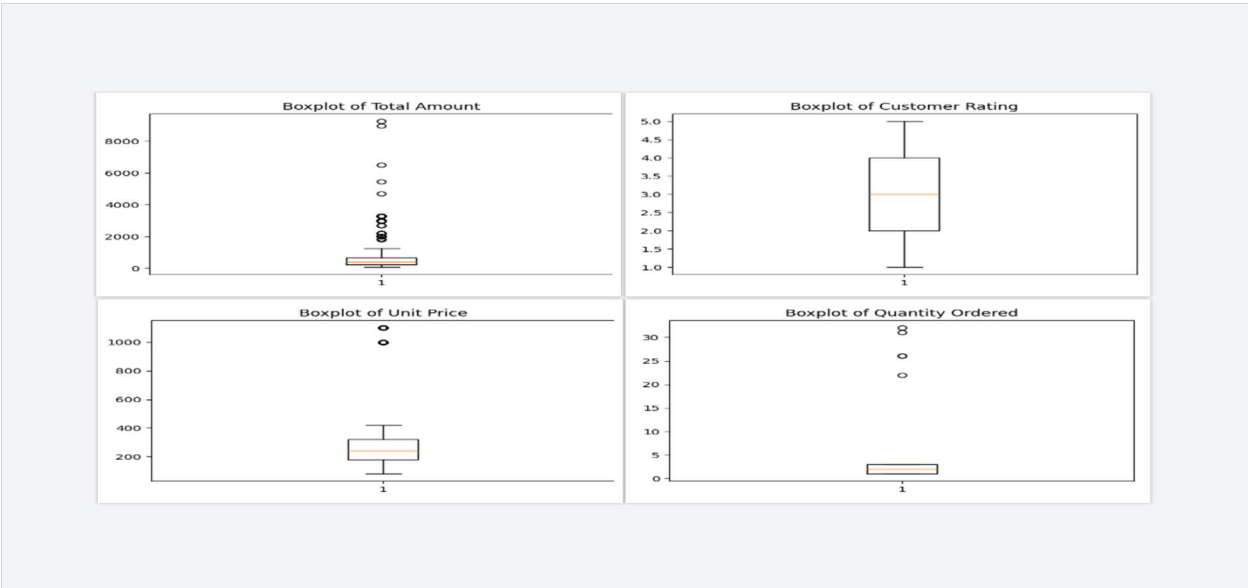
Statistic	Value (INR)	Note
Q1 (25th percentile)	₹255	Corrected from ₹280
Q3 (75th percentile)	₹660	—
IQR (Q3 – Q1)	₹405	Corrected from ₹380
Upper Bound	₹1,267.50	$Q3 + 1.5 \times \text{IQR}$
Lower Bound	-₹352.50	Ignored (no negatives)
Outlier Count	62 orders	> ₹1,267.50

Corrections: The original report stated Q1 = ₹280 and IQR = ₹380. The correct values are Q1 = ₹255 and IQR = ₹405 (Q3 = ₹660, both unchanged).

Top 3 Outlier Orders

Order ID	Date	Product	Qty	Total (INR)
ORD00698	2025-05-11	Veg Sandwich	22	₹9,240
ORD00700	2025-04-16	Pasta Salad	32	₹8,960
ORD00696	2025-04-18	Cold Brew	31	₹6,510

These are bulk / group orders. They were retained in the dataset as they represent genuine revenue from corporate or event orders. However, they pull the mean and standard deviation significantly upward; median-based metrics are more representative for typical customers.



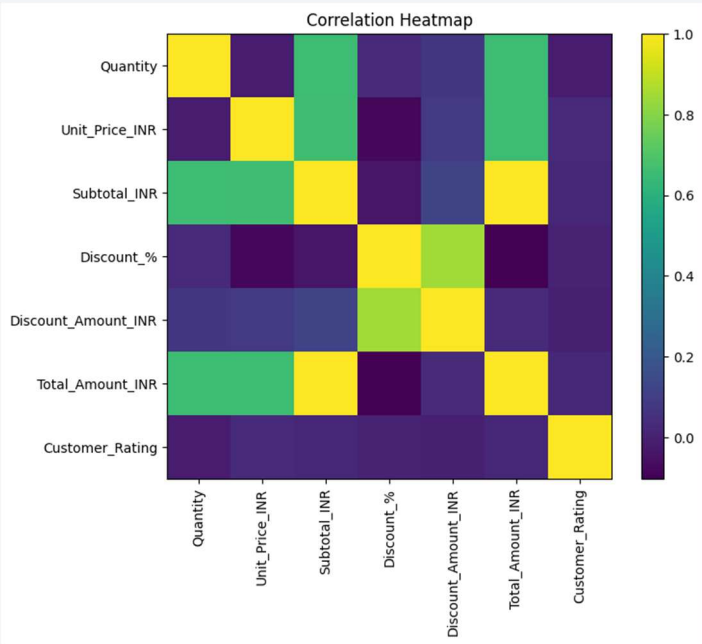
10. Correlation Analysis

Pearson correlation coefficients were computed across all numeric order columns. Key results:

Variable	Qty	Unit ₹	Subtotal	Disc %	Disc ₹	Total ₹	Rating
Quantity	1.00	0.00	0.66	0.01	0.24	0.66	-0.03
Unit Price ₹	0.00	1.00	0.62	0.01	0.37	0.66	0.02
Subtotal ₹	0.66	0.62	1.00	0.01	0.43	1.00	0.02
Discount %	0.01	0.01	0.01	1.00	0.85	-0.01	-0.02
Discount ₹	0.24	0.37	0.43	0.85	1.00	0.37	0.04
Total ₹	0.66	0.66	1.00	-0.01	0.37	1.00	0.02
Rating	-0.03	0.02	0.02	-0.02	0.04	0.02	1.00

Key Correlation Findings

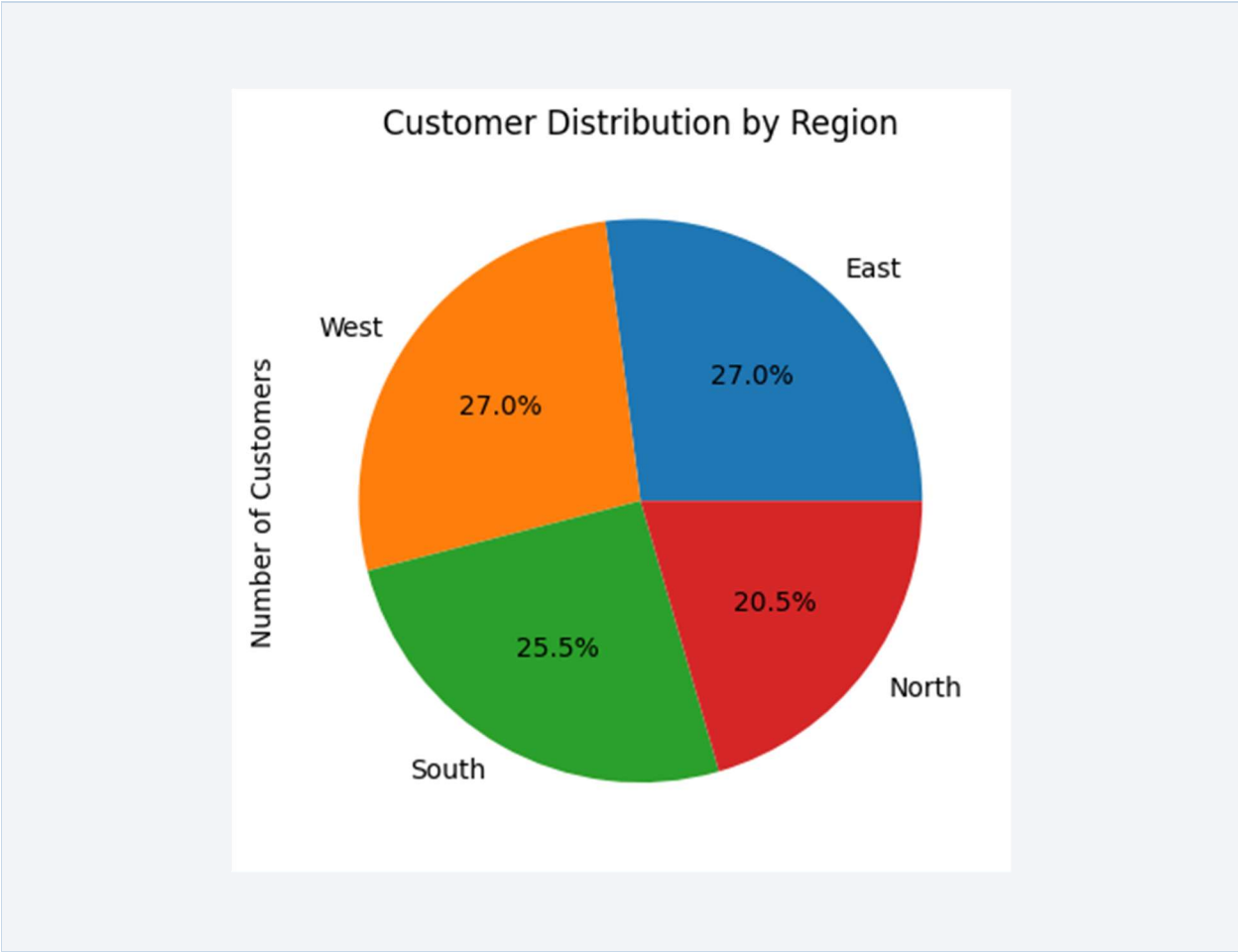
- Subtotal ↔ Total Amount: $r = 1.00$ — Expected (Total = Subtotal – Discount).
- Quantity ↔ Total Amount: $r = 0.66$ — More items ordered leads to higher spend.
- Unit Price ↔ Total Amount: $r = 0.66$ — Higher-priced products drive larger totals.
- Discount % ↔ Discount Amount: $r = 0.85$ — Strong and consistent relationship.
- Customer Rating ↔ Total Amount: $r \approx 0.02$ — Essentially zero correlation; spending levels do not influence ratings.



11. Customer Profile

Attribute	Detail
Total Customers	200
Gender Split	Female: 106 (53.0%) Male: 94 (47.0%)
Age Range	18 – 65 years 8 missing values
Top Cities	Kolkata (19), Bengaluru (15), Nagpur (12), Ahmedabad (12), Mumbai (12)
Regional Split	West: 54 East: 54 South: 51 North: 41
Membership Tiers	7 customers have no tier assigned (nulls)

Customers are distributed fairly evenly across regions (West and East tied at 54 each), with a slight female majority. All age values are within the expected adult range (18–65). The 7 null membership tier records should be resolved to enable tier-based campaign targeting.



12. Insights & Recommendations

12.1 Revenue & Sales Strategy

- Focus on high-value bulk orders: 62 outlier orders contributed disproportionately to Q2 revenue. Introduce corporate account programmes with volume pricing to formalise and grow this segment.
- Revenue is declining quarter-over-quarter (–15.3% from April to June). Investigate whether this is seasonal or structural and launch targeted June–July promotions to stabilise revenue.
- Dine-In is the dominant channel (44.4%). Invest in in-store ambience and loyalty touchpoints. Consider upselling meal combos to increase average basket size.

12.2 Category & Product Strategy

- Meals are the top revenue category (₹262,113). Expand meal combo deals and lunch specials to drive further growth.
- Merchandise contributes ₹234,673 — the second-highest category. Branded products (mugs, accessories) are clearly popular; consider seasonal or limited-edition launches.
- Delivery accounts for only 23.7% of revenue despite high digital ordering demand. Strengthen delivery partnerships and in-app promotions to capture this channel's potential.

12.3 Loyalty & Discount Strategy

- "10% Off Next Visit" is the strongest performer by revenue (₹133,449) and redemption (245). Increase the volume of this offer to more customers.
- "Double Loyalty Points" drives ₹94,270 in revenue — repeat-visit mechanics are highly effective. Introduce points expiry deadlines to create urgency.
- Discounted orders average ₹483 vs ₹599 for non-discounted. Ensure discounts include minimum spend thresholds (e.g., min ₹500) to protect margin.

12.4 Customer Experience & Data Quality

- Average customer rating is 3.05/5 — below a "good" threshold. Investigate root causes (delivery speed, product quality, incorrect orders) and establish a feedback response process.
- Ratings show no correlation with spend ($r \approx 0.02$), meaning all customer segments — low and high spend — are equally likely to leave low ratings. Service quality improvement must be universal, not targeted only at high spenders.
- Resolve the 7 null Membership_Tier records in Customer_Information to enable tier-based campaign targeting.
- The 2 null Total_Amount_INR records in Order_Information should be investigated and corrected or removed to maintain financial reporting accuracy.

Appendix A — Code Reference

SQL Examples

```
-- Count total orders and large orders
SELECT COUNT(*) AS total_orders,
       SUM(CASE WHEN Total_Amount_INR > 1000 THEN 1 ELSE 0 END) AS large_orders
FROM Order_Information;

-- Revenue by order type
SELECT Order_Type, SUM(Total_Amount_INR) AS revenue
FROM Order_Information GROUP BY Order_Type;

-- Loyalty offer redemptions
SELECT l.Reward_Offer, COUNT(*) AS offers_sent,
       SUM(CASE WHEN o.Reward_Applied = l.Reward_Offer THEN 1 ELSE 0 END) AS redeemed
FROM Loyalty_Campaign_Information l
LEFT JOIN Order_Information o ON l.Customer_ID = o.Customer_ID
GROUP BY l.Reward_Offer;
```

Python Reference

```
import pandas as pd
orders = pd.read_csv('Order_Information.csv')
menu = pd.read_csv('Menu_Information.csv')
loyalty = pd.read_csv('Loyalty_Campaign_Information.csv')
customers = pd.read_csv('Customer_Information.csv')

# Outlier detection (IQR method)
Q1, Q3 = orders['Total_Amount_INR'].quantile([0.25, 0.75])
IQR = Q3 - Q1 # 405
outliers = orders[orders['Total_Amount_INR'] > Q3 + 1.5*IQR]

# Monthly revenue
orders['Order_Date'] = pd.to_datetime(orders['Order_Date'])
monthly = orders.set_index('Order_Date').resample('ME')['Total_Amount_INR'].sum()

# Correlation matrix
cols = ['Quantity', 'Unit_Price_INR', 'Subtotal_INR', 'Discount_%',
        'Discount_Amount_INR', 'Total_Amount_INR', 'Customer_Rating']
corr = orders[cols].corr()
```